

SUMMARY OF PRODUCT CHARACTERISTICS

1 NAME OF THE MEDICINAL PRODUCT

Enstilar® 50 micrograms/g + 0.5 mg/g cutaneous foam

2 QUALITATIVE AND QUANTITATIVE COMPOSITION

One gram of cutaneous foam contains 50 micrograms of calcipotriol (as monohydrate) and 0.5 mg of betamethasone (as dipropionate).

Excipient with known effect:

Butylhydroxytoluene (E321) 50 micrograms/g cutaneous foam.

For the full list of excipients, see section 6.1.

3 PHARMACEUTICAL FORM

Cutaneous foam.

After spraying, a white to off-white foam is formed.

The foam has the appearance of a non-expanding foam that gradually collapses after spraying.

4 CLINICAL PARTICULARS

4.1 Therapeutic indications

Topical treatment of psoriasis vulgaris in adults.

4.2 Posology and method of administration

Posology

Flare treatment

Enstilar® foam should be applied to the affected area once daily. The recommended treatment period is 4 weeks. If it is necessary to continue or restart treatment after this period, treatment should be continued after medical review and under regular supervision.

Long-term maintenance treatment

Patients who have responded at 4 weeks' treatment using Enstilar once daily are suitable for long-term maintenance treatment. Enstilar should be applied twice weekly on two non-consecutive days to areas previously affected by psoriasis vulgaris. Between applications there should be 2-3 days without Enstilar treatment. If signs of a relapse occur, flare treatment, as described above, should be re-initiated.

Maximum dose

The daily maximum dose of Enstilar® should not exceed 15 g, i.e. one 60 g can should last for at least 4 days of treatment. 15 g corresponds to the amount administered from the can if the actuator is fully depressed for approximately one minute. A two-second application delivers approximately 0.5 g. As a guide, 0.5 g of foam should cover an area of skin roughly corresponding to the surface area of an adult hand.

If using other topical products containing calcipotriol in addition to Enstilar®, the total dose of all calcipotriol containing products should not exceed 15 g per day.

The total body surface area treated should not exceed 30%.

Special populations

Renal and hepatic impairment

The safety and efficacy of Enstilar® foam in patients with severe renal insufficiency or severe hepatic disorders have not been evaluated.

Paediatric population

The safety and efficacy of Enstilar® foam in children below 18 years have not been established. Currently available data in children aged 12 to 17 years are described in sections 4.8 and 5.1, but no recommendation on a posology can be made.

Method of administration

For cutaneous use.

The can should be shaken for a few seconds before use. Enstilar® should be applied by spraying holding the can at least 3 cm from the skin. The foam can be sprayed holding the can in any orientation except horizontally.

Enstilar® should be sprayed directly onto each affected skin area and rubbed in gently.

If used on the scalp, Enstilar should be sprayed into the palm of the hand and then applied to affected scalp areas with the fingertips. Hair washing instructions are provided in the package leaflet.

The hands should be washed after using Enstilar® (unless Enstilar® is used to treat the hands) to avoid accidentally spreading to other parts of the body as well as unintended drug absorption on the hands. Application under occlusive dressings should be avoided since it increases the systemic absorption of corticosteroids. It is recommended not to take a shower or bath immediately after application of Enstilar®. Let the foam remain on the scalp and/or skin during the night or during the day.

4.3 Contraindications

Hypersensitivity to the active substances or to any of the excipients listed in section 6.1.

Enstilar® is contraindicated in erythrodermic and pustular psoriasis.

Due to the content of calcipotriol, Enstilar® is contraindicated in patients with known disorders of calcium metabolism (see section 4.4).

Due to the content of corticosteroid, Enstilar® is contraindicated in the following conditions if present in the treatment area: viral (e.g. herpes or varicella) lesions of the skin, fungal or bacterial skin infections, parasitic infections, skin manifestations in relation to tuberculosis, perioral dermatitis, atrophic skin, striae atrophicae, fragility of skin veins, ichthyosis, acne vulgaris, acne rosacea, rosacea, ulcers, and wounds (see section 4.4).

4.4 Special warnings and precautions for use

Effects on endocrine system:

Adverse reactions found in connection with systemic corticosteroid treatment, such as adrenocortical suppression or impaired glycaemic control of diabetes mellitus may occur also during topical corticosteroid treatment due to systemic absorption.

Application under occlusive dressings should be avoided since it increases the systemic absorption of corticosteroids. Application on large areas of damaged skin, or on mucous

membranes or in skin folds should be avoided since it increases the systemic absorption of corticosteroids (see section 4.8).

Visual disturbance:

Visual disturbance may be reported with systemic and topical corticosteroid use. If a patient presents with symptoms such as blurred vision or other visual disturbances, the patient should be considered for a referral to an ophthalmologist for evaluation of possible causes which may include cataract, glaucoma or rare diseases such as central serous chorioretinopathy (CSCR) which have been reported after use of systemic and topical corticosteroids.

Effects on calcium metabolism:

Due to the content of calcipotriol in Enstilar®, hypercalcaemia may occur. Serum calcium is normalised when treatment is discontinued. The risk of hypercalcaemia is minimal when the maximum daily dose of Enstilar® (15 g) is not exceeded (see section 4.2).

Local adverse reactions:

Enstilar® contains a potent group III-steroid and concurrent treatment with other steroids on the same treatment area must be avoided.

The skin of the face and genitals is very sensitive to corticosteroids. The medicinal product should not be used in these areas.

The patient must be instructed in correct use of the product to avoid application and accidental transfer to the face, mouth, and eyes. Hands must be washed after each application to avoid accidental transfer to these areas as well as unintended drug absorption on the hands.

Concomitant skin infections:

If lesions become secondarily infected, they should be treated with antimicrobiological therapy. However, if infection worsens, treatment with corticosteroids should be discontinued (see section 4.3).

Discontinuation of treatment:

When treating psoriasis with topical corticosteroids, there may be a risk of rebound effects when discontinuing treatment. Medical supervision should therefore continue in the post-treatment period.

Long-term use:

Long-term use of corticosteroids may increase the risk of local and systemic adverse reactions. Treatment should be discontinued in case of adverse reactions related to long-term use of corticosteroid (see section 4.8).

Unevaluated use:

There is no experience with the use of Enstilar® in guttate psoriasis.

Adverse reactions to excipients:

Enstilar® contains butylhydroxytoluene (E321) as an excipient, which may cause local skin reactions (e.g. contact dermatitis) or irritation to the eyes and mucous membranes.

4.5 Interaction with other medicinal products and other forms of interaction

No interaction studies have been performed with Enstilar®.

4.6 Fertility, pregnancy and lactation

Pregnancy:

There are no adequate data from the use of Enstilar® in pregnant women. When administered orally in animals, studies of calcipotriol have not shown teratogenic effects, though reproductive toxicity has been shown (see section 5.3). Studies in animals with glucocorticoids have shown reproductive toxicity (see section 5.3), but a number of epidemiological studies (less than 300 pregnancy outcomes) have not revealed congenital anomalies among infants born to women treated with corticosteroids during pregnancy. The potential risk for humans is uncertain.

Therefore, during pregnancy, Enstilar® should only be used when the potential benefit justifies the potential risk.

Breast-feeding:

Betamethasone passes into breast milk, but risk of an adverse reaction in the infant is very small with therapeutic doses. There are no data on the excretion of calcipotriol in breast milk. Caution should be exercised when prescribing Enstilar® to women who breast-feed. The patient should be instructed not to use Enstilar® on the breast when breast-feeding.

Fertility:

Studies in rats with oral doses of calcipotriol or betamethasone dipropionate demonstrated no impairment of male and female fertility (see section 5.3). There are no data on human fertility,

4.7 Effects on ability to drive and use machines

Enstilar® has no or negligible influence on the ability to drive and to use machines.

4.8 Undesirable effects

The estimation of the frequency of adverse reactions is based on a pooled analysis of data from clinical studies.

The most frequently reported adverse reactions during treatment are application site reactions.

Adverse reactions are listed by MedDRA SOC and the individual adverse reactions are listed starting with the most frequently reported. Within each frequency grouping, adverse reactions are presented in the order of decreasing seriousness.

Very common ($\geq 1/10$)

Common ($\geq 1/100$ to $< 1/10$)

Uncommon ($\geq 1/1,000$ to $< 1/100$)

Rare ($\geq 1/10,000$ to $< 1/1,000$)

Very rare ($< 1/10,000$)

Not known (cannot be estimated from the available data)

Infections and infestations	
Uncommon $\geq 1/1,000$ to $< 1/100$	Folliculitis
Immune system disorders	
Uncommon $\geq 1/1,000$ to $< 1/100$	Hypersensitivity
Metabolism and nutrition disorders	
Uncommon $\geq 1/1,000$ to $< 1/100$	Hypercalcaemia*
Eye disorders	

Not known	Vision, blurred**
Skin and subcutaneous tissue disorders	
Uncommon $\geq 1/1,000$ to $< 1/100$	Skin hypopigmentation
Not known	Hair colour changes***
General disorders and administration site conditions	
Uncommon $\geq 1/1,000$ to $< 1/100$	Rebound effect Application site pruritus Application site irritation Application site pain****
Not known	Application site erythema*****

* Mild hypercalcaemia has been observed.

** See section 4.4.

*** Transient discolouration of the hair at scalp application site, to a yellowish colour in white or grey hair, has been reported for calcipotriol and betamethasone combination products.

**** Application site burning is included in application site pain.

***** Based on post-marketing experience.

Paediatric population

No clinically relevant differences between the safety profiles in adult and adolescent populations have been observed. A total of 106 adolescent subjects were treated in one open-label clinical trial. See section 5.1 for further details regarding this trial.

The following adverse reactions are considered to be related to the pharmacological classes of calcipotriol and betamethasone, respectively:

Calcipotriol:

Adverse reactions include application site reactions, pruritus, skin irritation, burning and stinging sensation, dry skin, erythema, rash, dermatitis, psoriasis aggravated, photosensitivity and hypersensitivity reactions, including very rare cases of angioedema and facial oedema.

Systemic effects after topical use may appear very rarely causing hypercalcaemia or hypercalciuria (see section 4.4).

Betamethasone (as dipropionate):

Local reactions can occur after topical use, especially during prolonged application, including skin atrophy, telangiectasia, striae, folliculitis, hypertrichosis, perioral dermatitis, allergic contact dermatitis, depigmentation, and colloid milia.

When treating psoriasis with topical corticosteroids, there may be a risk of generalised pustular psoriasis.

Systemic reactions due to topical use of corticosteroids are rare in adults; however, they can be severe. Adrenocortical suppression, cataract, infections, impaired glycaemic control of diabetes mellitus, and increase of intra-ocular pressure can occur, especially after long-term treatment. Systemic reactions occur more frequently when applied under occlusion (plastic, skin folds), when applied onto large areas, and during long-term treatment (see section 4.4).

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via HPRA Pharmacovigilance, website: www.hpra.ie

4.9 Overdose

Usage above the recommended dose may cause elevated serum calcium which subsides when treatment is discontinued. The symptoms of hypercalcaemia include polyuria, constipation, muscle weakness, confusion, and coma.

Excessive prolonged use of topical corticosteroids may result in adrenocortical suppression which is usually reversible. Symptomatic treatment may be indicated.

In case of chronic toxicity the corticosteroid treatment must be discontinued gradually.

5 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antipsoriatics. Other antipsoriatics for topical use, Calcipotriol, combinations. ATC Code: D05AX52.

Mechanism of action:

Enstilar® foam combines the pharmacological effects of calcipotriol hydrate as a synthetic vitamin D3 analogue and betamethasone dipropionate as a synthetic corticosteroid.

In psoriasis, vitamin D and its analogues act mainly to inhibit keratinocyte proliferation and induce keratinocyte differentiation. The underlying antiproliferative mechanism of vitamin D in keratinocytes involves the induction of the growth inhibitory factor transforming growth factor- β and of cyclin-dependent kinase inhibitors, with subsequent growth arrest in the G1 phase of the cell cycle plus down-regulation of the two proliferation factors early growth response-1 and polo-like kinase-2.

In addition, vitamin D has an immunomodulatory effect, suppressing activation and differentiation of Th17/Th1 cells while inducing a Th2/Treg response.

In psoriasis, corticosteroids suppress the immune system, particularly pro-inflammatory cytokines and chemokines, thereby inhibiting T-cell activation. At the molecular level, corticosteroids act via the intracellular glucocorticoid receptor and the anti-inflammatory function is due to transrepression of pro-inflammatory transcription factors such as nuclear factor κ B, activator protein-1, and interferon regulatory factor-3.

In combination, calcipotriol monohydrate and betamethasone dipropionate promote greater anti-inflammatory and anti-proliferative effects than either component alone.

Pharmacodynamic effects:

Short term data

Under maximum use conditions, in subjects with extensive psoriasis on the body and scalp treated for up to 4 weeks, adrenal response to ACTH was determined by measuring serum cortisol levels. None of 35 subjects had suppressed serum cortisol levels at 30 or 60 minutes post ACTH stimulation. Thus it appears that for Enstilar®, the risk of adrenal suppression is low when applied to extensive psoriasis vulgaris for 4 weeks. Similarly, there was no indication

of abnormal calcium metabolism following application of Enstilar® to extensive psoriasis vulgaris for 4 weeks.

Long-term data

The adrenal response to ACTH challenge was evaluated in adult subjects with moderate to severe psoriasis vulgaris involving at least 10% of the body surface area. The subjects were randomised to receive Enstilar or foam vehicle twice weekly for up to 52 weeks (long-term maintenance treatment). Subjects experiencing a relapse were treated with Enstilar once daily for 4 weeks, then continued randomised treatment.

The trial results were in line with a low risk of adrenal suppression in subjects with extensive psoriasis (BSA 10-30%) who use Enstilar twice weekly and as outlined for up to 52 weeks. There was no clinically relevant effect on the calcium metabolism in this trial.

Clinical efficacy:

Short-term data

The efficacy of once daily use of Enstilar® has been investigated in three randomised, double-blind or investigator-blind, 4-week clinical trials including more than 1,100 subjects with psoriasis on the body (also scalp in Trial Two) of at least mild severity according to the Physician's Global Assessment of disease severity (PGA), affecting at least 2% body surface area (BSA), and with a modified Psoriasis Area Severity Index (m-PASI) of at least 2. The physician's global assessment is made using a 5-point scale (clear, almost clear, mild, moderate, and severe) based on the average psoriatic lesion.

The primary endpoint was subjects with 'treatment success' ('clear' or 'almost clear' for subjects with at least moderate disease at baseline, 'clear' for subjects with mild disease at baseline) according to the PGA at Week 4.

Disease-related baseline characteristics

	Trial One (N=426)	Trial Two (N=302)	Trial Three (N=376)
Baseline disease severity (PGA):			
Mild	65 (15.3%)	41 (13.6%)	63 (16.8%)
Moderate	319 (74.9%)	230 (76.2%)	292 (77.7%)
Severe	42 (9.9%)	31 (10.3%)	21 (5.6%)
Mean BSA (range)	7.5% (2-30%)	7.1% (2-28%)	7.5% (2-30%)
Mean m-PASI (range)	7.5 (2.0-47.0)	7.6 (2.0-28.0)	6.8 (2.0-22.6)

Percentage of subjects with 'treatment success' according to the PGA of the body at Week 4

	Enstilar®	Foam vehicle	BDP in foam vehicle	Calcipotriol in foam vehicle	Daivobet Ointment	Ointment vehicle
Trial One	(N=323)	(N=103)	-	-	-	-

	53.3%	4.8%				
Trial Two	(N=100) 45.0%	-	(N=101) 30.7%	(N=101) 14.9%	-	-
Trial Three	(N=100) 54.6%	(N=49) 6.1%	-	-	(N=135) 43.0%	(N=145) 7.8%

Results for the primary endpoint ‘treatment success’ (PGA) of body at Week 4 showed Enstilar® to be statistically significantly more effective than all the comparators included and responses were observed in all categories of baseline disease severity.

In Trial Two, the effect of Enstilar® on scalp psoriasis was investigated as the percentage of subjects with ‘treatment success’ according to the PGA of the scalp at Week 4.

Percentage of subjects with ‘treatment success’ according to the PGA of the scalp at Week 4

	Enstilar®	BDP in foam vehicle	Calcipotriol in foam vehicle
Trial Two	(N=100) 53.0 %	(N=101) 47.5 %	(N=101) 35.6 %

Enstilar® was statistically significantly more effective compared to calcipotriol and also associated with a higher rate of treatment success than BDP but this comparison did not reach statistical significance.

The effect of Enstilar® on itch and itch-related sleep loss was investigated in Trial One using a visual analogue scale (VAS) ranging from 0 mm (no itch/no sleep loss at all) to 100 mm (worst itch you can imagine/worst possible sleep loss). A statistically significantly higher number of subjects in the Enstilar® group compared to vehicle achieved a 70% reduction in itch and itch-related sleep loss from Day 3 and throughout the treatment period.

The effect of Enstilar® on quality of life was investigated in Trial One using the generic EQ-5D-5L questionnaire and the dermatologically specific DLQI questionnaire. Statistically significantly greater improvement in quality of life in favour of Enstilar® was demonstrated for DLQI from Week 1 and throughout the treatment period and for EQ-5D-5L at Week 4.

Long-term data

The efficacy and safety of treatment with Enstilar was investigated in a randomised, double-blind vehicle-controlled trial (Trial Four). Subjects were treated once daily with open-label Enstilar for 4 weeks and responders were then randomised to receive Enstilar (long-term maintenance treatment) or foam vehicle twice weekly for up to 52 weeks. Subjects in both treatment arms experiencing a relapse were treated once daily with Enstilar for 4 weeks, and those responding then continued randomized treatment.

Disease-related baseline characteristics (all randomised subjects)

	Trial Four (N=545)
Baseline disease severity (PGA)	
Mild	58 (10.6%)

Moderate	447 (82.0%)
Severe	40 (7.3%)
Mean BSA (range)	8.3 (1.0-38.0)
Mean m-PASI (range)	7.8 (2.0-28.0)

Subjects on long-term maintenance treatment with Enstilar had longer time to first relapse, greater proportion of days in remission during the trial, and fewer relapses than subjects using foam vehicle. The table below presents an overview of the effect on relapse in this trial.

Summary of efficacy up to 52 weeks of long-term maintenance treatment (Trial Four)

Endpoint	Observed data in the trial		Statistical analysis results (N=521)*	
	Long-term maintenance + relapse treatment (N=256)	Vehicle + relapse treatment (N=265)	Estimates [95% CI]	p-value
Primary: Time to first relapse	Median time to first relapse=56 days	Median time to first relapse=30 days	HR=0.57 [0.47; 0.69] (Reduction of 43% [31%; 53%])	p<0.001
Secondary: Proportion of days in remission	Median proportion of days=69.3%	Median proportion of days=56.6%	DP=11% [8%;14%] (Increase of 41 [29; 53] days)	p<0.001
Secondary: Number of relapses	Median number of relapses=2.0	Median number of relapses=3.0	RR=0.54 [0.46;0.63] (Reduction of 46% [37%; 54%])	p<0.001

*Statistical analysis compared long-term maintenance treatment + relapse treatment with Vehicle + relapse treatment

CI: Confidence interval; DP: Difference in proportion of days per year; HR: Hazard-ratio;

N: number of subjects in full analysis set; RR: Rate-ratio

Paediatric population

The effects on calcium metabolism were investigated in an uncontrolled, open-label, 4-week trial in 106 adolescents aged 12 to 17 years with scalp and body psoriasis. The subjects used up to 105 g Enstilar per week. No cases of hypercalcaemia and no clinically relevant changes in urinary calcium were reported.

The adrenal response to ACTH challenge was measured in a subset of 33 subjects with extensive plaque psoriasis involving at least 20% of the scalp and 10% of the body surface area. After 4 weeks of treatment with Enstilar, 2 subjects had a cortisol level ≤ 18 mcg/dL at 30 minutes after ACTH challenge, but had normal response at 60 minutes. A third subject had minimal cortisol response to the ACTH challenge test at baseline resulting in inconclusive results after the treatment. None of these cases had any clinical manifestations.

5.2 Pharmacokinetic properties

Following systemic exposure, both active ingredients – calcipotriol and betamethasone dipropionate – are rapidly and extensively metabolised.

The main route of excretion of calcipotriol is via faeces (rats and minipigs) and for betamethasone dipropionate it is via urine (rats and mice). In rats, tissue distribution studies with radiolabelled calcipotriol and betamethasone dipropionate showed that the kidney and liver had the highest level of radioactivity.

The extent of percutaneous absorption of the two active ingredients following topical application of Enstilar® was determined in the HPA axis trial in subjects with extensive psoriasis vulgaris (see section 5.1). Calcipotriol and betamethasone dipropionate were below the lower limit of quantification in most samples from 35 patients treated once daily for 4 weeks for extensive psoriasis involving the body and scalp. Calcipotriol was quantifiable at some time point in 1 subject, betamethasone dipropionate in 5 subjects and metabolites of calcipotriol and betamethasone dipropionate were detectable in 3 and 27 subjects, respectively.

5.3 Preclinical safety data

Studies of corticosteroids in animals have shown reproductive toxicity (cleft palate, skeletal malformations). In reproduction toxicity studies with long-term oral administration of corticosteroids to rats, prolonged gestation and prolonged and difficult labour were detected. Moreover, reduction in offspring survival, body weight and body weight gain was observed. There was no impairment of fertility. The relevance for humans is unknown.

Calcipotriol has shown maternal and foetal toxicity in rats and rabbits when given by the oral route at doses of 54 µg/kg/day and 12 µg/kg/day, respectively. The foetal abnormalities observed with concomitant maternal toxicity included signs indicative of skeletal immaturity (incomplete ossification of the pubic bones and forelimb phalanges, and enlarged fontanelles) and an increased incidence of supernumerary ribs.

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity and genotoxicity.

Dermal carcinogenicity studies with calcipotriol and betamethasone dipropionate in mice and oral carcinogenicity study in rats revealed no special risk to humans.

In a local tolerability study in minipigs, Enstilar® caused mild to moderate skin irritation.

6 PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Liquid paraffin

Polyoxypropylene stearyl ether

All-rac- α -tocopherol

White soft paraffin

Butylhydroxytoluene (E321)

Butane

Dimethyl ether

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

2 years.
6 months after first opening

6.4 Special precautions for storage

Do not store above 25°C.

Caution when Handling:
Extremely flammable aerosol.

Pressurised container: May burst if heated.

Protect from sunlight.

Do not expose to temperatures exceeding 50°C.

Do not pierce or burn, even after use.

Do not spray on an open flame or other ignition source.

Keep away from sparks, open flames and other ignition sources.

No smoking.

6.5 Nature and contents of container

Aluminium can with a polyamide-imide inner lacquer, equipped with a continuous valve and actuator.

The can contains 60 g of foam, not including the amount of propellants.

Pack sizes: 60 g and 2 x 60 g

Not all pack sizes may be marketed.

7 MANUFACTURER

LEO Laboratories Ltd, 285, Cashel Road, Crumlin, Dublin 12, Ireland

8 DATE OF REVISION OF THE TEXT

December 2022